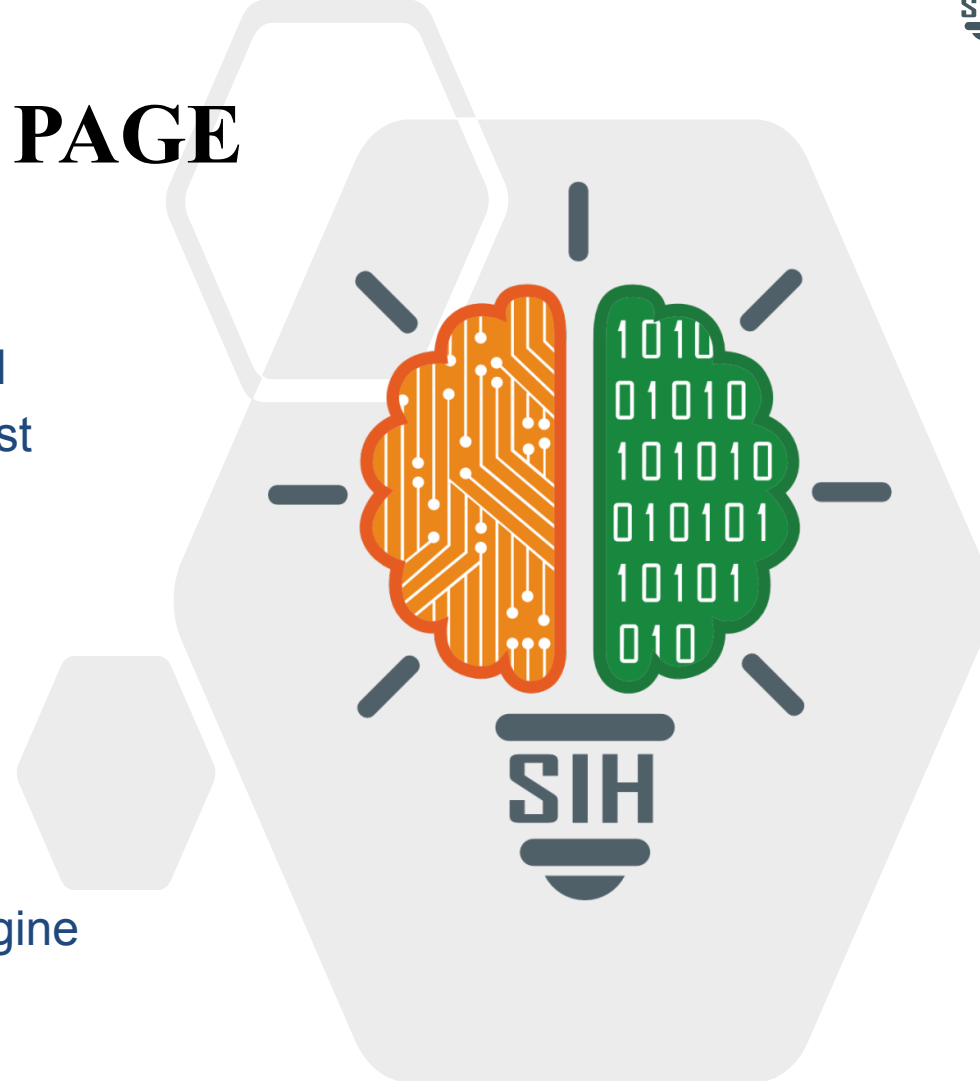


TITLE PAGE

- **Problem Statement ID** – SIH1431
- **Problem Statement Title** – Online personalized learning remediation/tutoring tool. Search for best teacher for specific topics.
- **Theme** – Smart Education
- **PS Category** – Software
- **Team ID** – [Registered Team ID]
- **Team Name** – **AUTHENTIX**
- **Proposed Solution** – VIDYA AI: Cognitive Remediation & Semantic Teacher Discovery Engine



VIDYA AI: Cognitive Remediation & Teacher Discovery

- **Closed-Loop Cognitive Architecture:** Integrates real-time student misconception diagnosis with semantic, topic-specific teacher discovery.
- **Knowledge DAG Tracing:** Traverses a Prerequisite Knowledge DAG to isolate the exact atomic barrier (e.g. Eigenvalue trace rule, Recursion base-cases) behind failed questions.
- **Deterministic Step-by-Step Proofs:** Instant KaTeX derivations backed by automated mathematical sanity assertions with zero hallucination.
- **On-Demand Micro-Tutoring:** Eliminates predatory ₹50,000 annual packages by offering 10-minute 1-on-1 peer doubt clearing sessions for pennies.
- **Cognitive Memory Twin:** SuperMemo SM-2 spaced repetition scheduler reinforces concepts before decay, lifting 30-day retention from 20% to >75%.

⚡ COGNITIVE REMEDIATION WORKFLOW

1. Student Doubt / Diagnostic Input

Formula OCR + AICTE / CBSE syllabus curriculum mapping

INPUT

▼ Misconception Detection

2. Prerequisite Knowledge DAG Traversal

Deep Knowledge Tracing isolates root atomic barrier

DIAGNOSIS

▼ Dual-Track Remediation

Track A: Instant AI

KaTeX proof + sanity check (<30s)

AI

Track B: Live Peer

10-min WebRTC micro-session

LIVE

▼ Mastery Verification

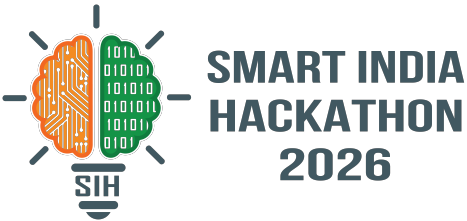
3. Adaptive Quiz & SM-2 Memory Twin

2-min verification quiz + automated 30-day spaced recall

MASTERY



TECHNICAL APPROACH



Technology Stack & Implementation Architecture

- **Frontend & Interaction:** React 19 PWA, TypeScript, Tailwind CSS, KaTeX LaTeX math parser, and interactive HTML5 canvas whiteboard.
- **AI & Diagnostics Core:** Deep Knowledge Tracing (DKT), Sentence-Transformers (all-MiniLM-L6-v2) for dense topic embeddings, and Gemini 1.5 Flash.
- **Vector DB & Data Tier:** PostgreSQL with pgvector (HNSW indexing for sub-15ms semantic searches), Redis session cache, and syllabus DAG.
- **Zero-Cost Live Media:** Peer-to-Peer WebRTC audio/video mesh and synchronized whiteboard with zero central relay bandwidth overhead.
- **Deterministic Assertions:** Automated algebraic consistency checks prevent model hallucinations before rendering to the student.

IMPLEMENTATION & DATA PIPELINE

TIER 1: CLIENT & PRESENTATION

Frontend

React 19 PWA + TypeScript + Tailwind CSS + KaTeX Math
Responsive Web/Mobile, OCR equation capture, Real-time Whiteboard

▼ (HTTPS / WebSockets REST API)

TIER 2: AI DIAGNOSTIC & MATCHING CORE

FastAPI

Knowledge DAG Tracing + Sentence-Transformers Embeddings
Sub-15ms semantic matching, Topic Affinity formula calculation (St)

▼ (SQL Queries & Vector Cosine Similarity)

TIER 3: HIGH-PERFORMANCE VECTOR STORAGE

Database

PostgreSQL + pgvector (HNSW Indexing) + Redis Cache
500,000+ Teacher topic vectors, Curriculum taxonomy, Session cache

▼ (Direct Peer Mesh & Micro-Payments)

TIER 4: LIVE REMEDIATION & MEMORY TWIN

WebRTC

P2P WebRTC Micro-Rooms + SuperMemo SM-2 Spaced Recall
Zero server media bandwidth cost, Automated 30-day concept review scheduler

Feasibility Analysis & Risk Mitigation

- **High 36-Hour Feasibility:** Core UI, AICTE R25 syllabus database, KaTeX math renderer, and quiz engine are already built and tested.
- **Zero Cost Burden:** 100% open-source software (HuggingFace embeddings, PostgreSQL pgvector, WebRTC) ensures free access without server costs.
- **Hardware Inclusivity:** Ultra-lightweight Progressive Web App (PWA) operates smoothly on entry-level Android devices and 2G/3G networks.
- **Risk 1 & Mitigation (AI Hallucinations):** Automated deterministic math sanity assertions (e.g. $\text{Trace} = \text{Sum of Eigenvalues}$) verify steps.
- **Risk 2 & Mitigation (Tutor Fraud):** Topic badges are tied strictly to objective student quiz pass rates, preventing fake reviews.
- **Risk 3 & Mitigation (Scalability):** Direct P2P WebRTC architecture handles 100,000+ simultaneous sessions without media server bandwidth costs.

36-HOUR HACKATHON DELIVERY ROADMAP

PHASE 1: FOUNDATION (READY)

0h – 6h

React 19 PWA, AICTE R25 Syllabus DB & KaTeX Engine

Pre-built responsive UI, dynamic equation rendering, topic taxonomy



PHASE 2: CORE VECTOR MATCHING

6h – 18h

PostgreSQL pgvector & Sentence-Transformers Integration

HNSW indexing, sub-15ms semantic search, teacher topic affinity score



PHASE 3: LIVE REMEDIATION & WEBRTC

18h – 28h

P2P Audio/Video Mesh & Collaborative Whiteboard

Signaling server, peer micro-rooms, KaTeX proof generator assertions



PHASE 4: TESTING & DEMO HARDENING

28h – 36h

Adaptive Quiz Loop, SM-2 Twin & Live Pitch Simulation

Zero-hallucination sanity checks, end-to-end user flow verification

95%**Faster Resolution**

<30s (AI) & <10m (Teacher) vs 24-48h

90%**Cost Reduction**

10-min micro-slots vs ₹50k bundles

90%**Time Saved**

Direct concept proof vs 45m YouTube search

3.75x**Retention Gain**

>75% 30-day recall with SM-2 Twin vs 20%

Demographic & Socio-Economic Reach

- **30+ Crore Students:** Democratizes elite tier-1 personalized tutoring for rural and government school learners across India.
- **Vernacular Pedagogy:** Multi-lingual remediation in Hindi, Bengali, Tamil, Telugu, and Marathi ensures linguistic inclusion.
- **Economic Liberation:** Replaces predatory ₹50,000 annual course packages with free AI derivations and ₹10 micro-tutoring slots.
- **Confidence Restoration:** Immediate root-cause diagnosis prevents academic dropouts by turning fear of subjects into mastery.

National Policy Alignment (NEP 2020 & SDGs)

- **NEP 2020 Section 4.3:** Directly implements experiential, adaptive learning and formative competency-based assessments.
- **NEP 2020 Section 23:** Champions educational technology integration and AI-driven personalized remedial support.
- **PARAKH NCERT Compliance:** Bridges learning gaps mapped against national performance standards and AICTE guidelines.
- **UN SDG 4 & 10:** Furthers Quality Education and Reduced Inequalities through equitable, accessible learning technology.

Academic Research & Algorithmic Foundations

- **Deep Knowledge Tracing (DKT):** Piech, C., Bassen, J., et al. (Stanford University), NeurIPS. Recurrent neural networks for modeling student knowledge over time.
- **Cognitive Memory & Spaced Repetition:** Wozniak, P., Optimization of Learning: Two Components of Long-Term Memory (SuperMemo SM-2 Algorithm).
- **Dense Semantic Embeddings:** Reimers, N. & Gurevych, I., Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks (EMNLP).
- **Bloom's 2-Sigma Problem:** Bloom, B. S., The 2 Sigma Problem: The Search for Methods of Group Instruction as Effective as One-to-One Tutoring.
- **Knowledge Graph Remediation:** Chen, P., et al., Prerequisite Concept Graphs for Automated Adaptive Learning Path Generation.

Government Directives & Curriculum Benchmarks

- **National Education Policy (NEP 2020):** Ministry of Education, Government of India. Sections 4.3 & 23 on digital pedagogy and adaptive remediation.
- **AICTE Model Curriculum (R25 Guidelines):** Higher technical education engineering syllabus taxonomy and course outcome standards.
- **PARAKH NCERT Standards:** National Assessment Centre guidelines on student diagnostics, holistic progress cards, and remedial feedback loops.
- **AISHE Survey (Ministry of Education):** Baseline data on student-to-faculty ratios (24:1 national average) justifying automated peer tutoring.
- **DIKSHA & SWAYAM PRABHA:** National digital infrastructure interoperability frameworks for open digital educational content.